PLTW POE

Activity 3.1.3 Series vs. Parallel Circuits

PLTW

SERIES VS. PARALLEL CIRCUITS

Measuring Voltage

GOALS	MATERIALS	RESOURCES

- Differentiate between series and parallel circuits.
- Solve for voltage, current, and/or resistance in series and parallel circuits.

GOALS	MATERIALS	RESOURCES

- Variable power supply
- Breadboard
- 100 Ω resistors
- 220 Ω resistors
- 330 Ω resistors
- Hookup wires

- Digital multimeter

GOALS | MATERIALS | RESOURCES

- [Series vs. Parallel Circuits (Microsoft Excel)]
- [Series vs. Parallel Circuits (Google Sheets)]
- [Activity 3.1.3 Series vs. Parallel Circuits (Downloadable PDF)]

In Activity 3.1.2 Ohm's Law Lab, you assembled several series circuits using different resistors. You also discovered that the current through a series circuit remained the same no matter where it was measured. Then, you learned that the total circuit resistance is equal to the sum of the resistance values of the individual components.

In this activity, you will use Autodesk® Tinkercad® to model both series and parallel circuits and determine whether the mathematical relationships you identified in Activity 3.1.2 Ohm's Law Lab still apply.

Series Circuits

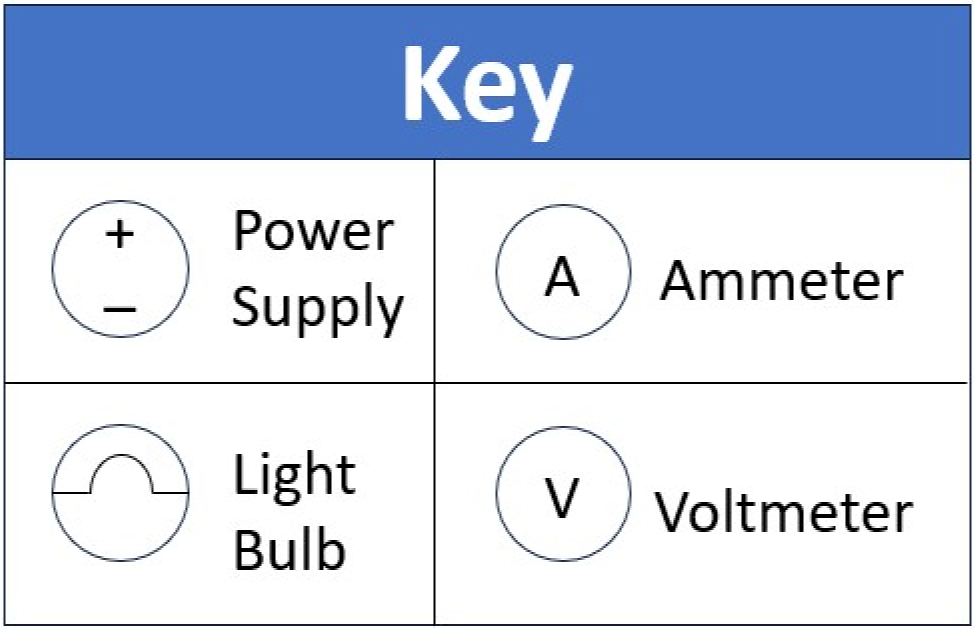
In this part of the activity, you will model series circuits with one bulb, two bulbs, and three bulbs. For each circuit, you will observe the individual bulb brightness. You will use multimeters to measure the current before each bulb and the voltage across each bulb.

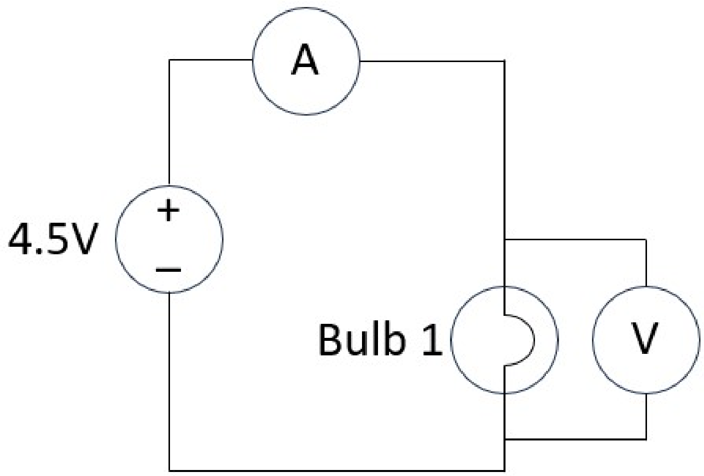
Turn and Talk

- How do you think adding bulbs to a series circuit will affect the brightness of each bulb?
- Predict any trends you might observe with respect to the voltage across the bulbs with each circuit. Explain your thinking.

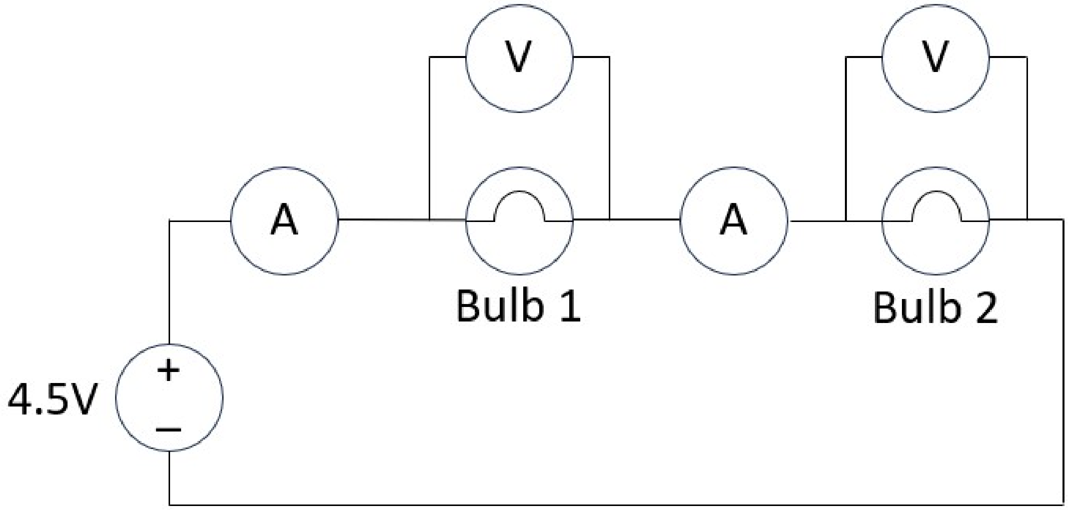
1. Download and save a copy of **Series vs. Parallel Circuits (Microsoft Excel)** or **Series vs. Parallel Circuits (Google Sheets)**. Open the Series Simulation tab.

2. Observe the circuit schematic key and the three circuit schematic figures below. For each one, note the value of the power supply (in volts) and the locations of multimeters to measure current (A) and voltage (V).

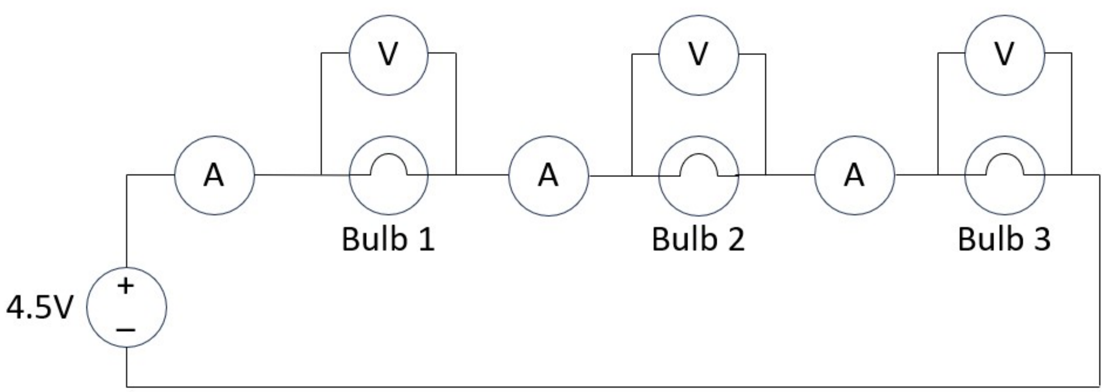

Key
+
–
Power Supply
A
Ammeter
Light Bulb
V
Voltmeter



CIRCUIT SCHEMATIC KEY | CIRCUIT SCHEMATIC 1 | CIRCUIT SCHEMATIC 2 | CIRCUIT SCHEMATIC 3



CIRCUIT SCHEMATIC KEY	CIRCUIT SCHEMATIC 1	CIRCUIT SCHEMATIC 2	CIRCUIT SCHEMATIC 3



3. Using the schematics from the figures above, model all three circuits in the same Tinkercad workspace.

4. Start the simulation in Tinkercad. Collect and record the data requested on the spreadsheet.

Reflection

- Referencing your data, how does the number of bulbs in a series circuit affect the current in that circuit? How does the number of bulbs affect the brightness of each bulb?
- How does the current you measured before each bulb relate to the total current measured next to the power supply?
- How does the total voltage that the power supply provided relate to the voltage across each bulb? Explain your thinking using data.

Mathematical Relationships for Series Circuits

Collecting experimental data can help you develop mathematical relationships (like *V = I x R*) that you can later apply to other situations under the same conditions. Using the data you collected, develop relationships for total current and voltage in a series circuit.

5. Describe and record the mathematical relationship between total current (I_T) and current through individual circuit components (I_1, I_2, I_3, ... I_n) in a series circuit.

6. Describe and record the mathematical relationship between total voltage (V_T) and voltage across individual circuit components (V_1, V_2, V_3, ... V_n) in a series circuit.

Time to Practice

Using your mathematical relationships, solve the following practice problems. You may need to algebraically manipulate your mathematical relationship depending on what quantity you are solving for.

7. Determine the voltage across R2 in Figure 1.

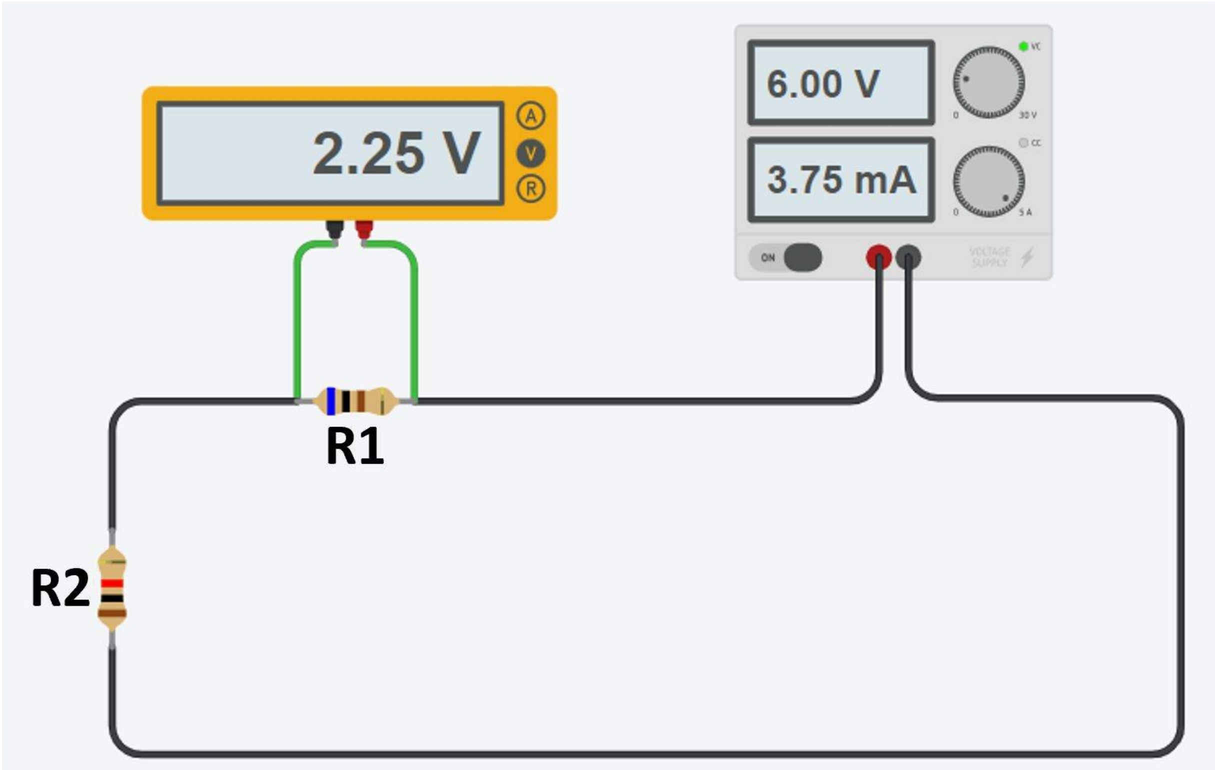


Figure 1. Series Circuit with Two Resistors

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8. A series circuit with two resistors is connected to a 9 V power supply. Determine the voltage across the second resistor if the voltage across one resistor is 4.5 V.

9. Determine the voltage across R3 in Figure 2.

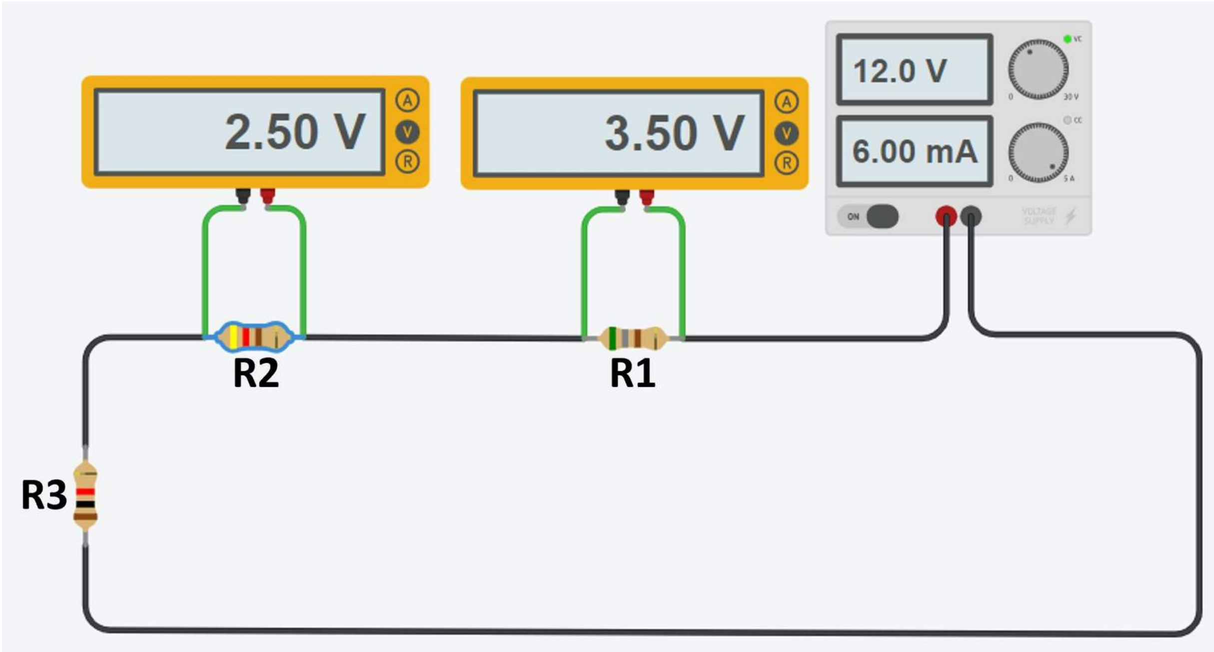


Figure 2. Series Circuit with Three Resistors

Parallel Circuits

So far in this activity, you have experimented exclusively with series circuits, or circuits with only one path for current to flow. You may also be familiar with parallel circuits, which have multiple paths for current to flow.

Figure 3 shows a series circuit with a power supply and two resistors with only one path for current to flow.

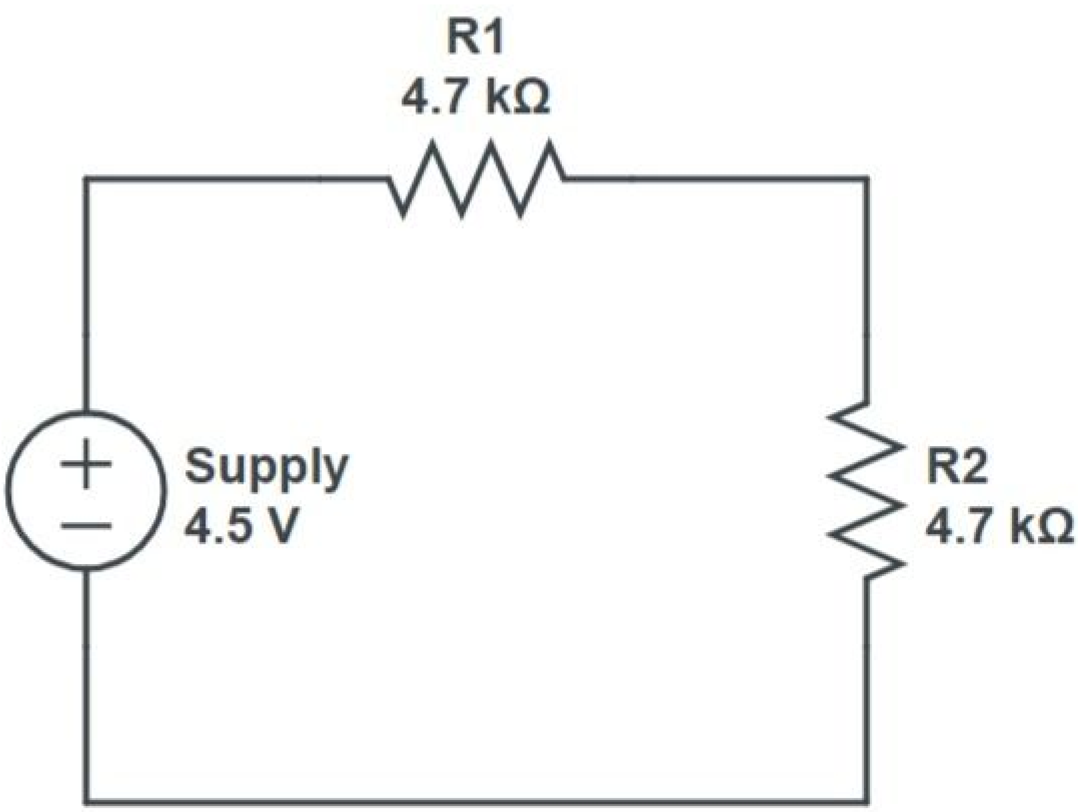


Figure 3. Series Circuit with Two Resistors

10

Draw a schematic diagram for a circuit with the same components as shown in Figure 3, but with two paths for current to flow.

Reflection

What are the advantages and disadvantages of your new parallel circuit over the circuit shown in Figure 3?

The schematic you drew is an example of a parallel circuit or a circuit with multiple paths for current to flow. In this part of the activity, you will use Tinkercad® to model parallel circuits with one bulb, two bulbs, and three bulbs. For each circuit, you will observe the individual bulb brightness, and measure the current before each bulb and the voltage across each bulb using multimeters.

Turn and Talk

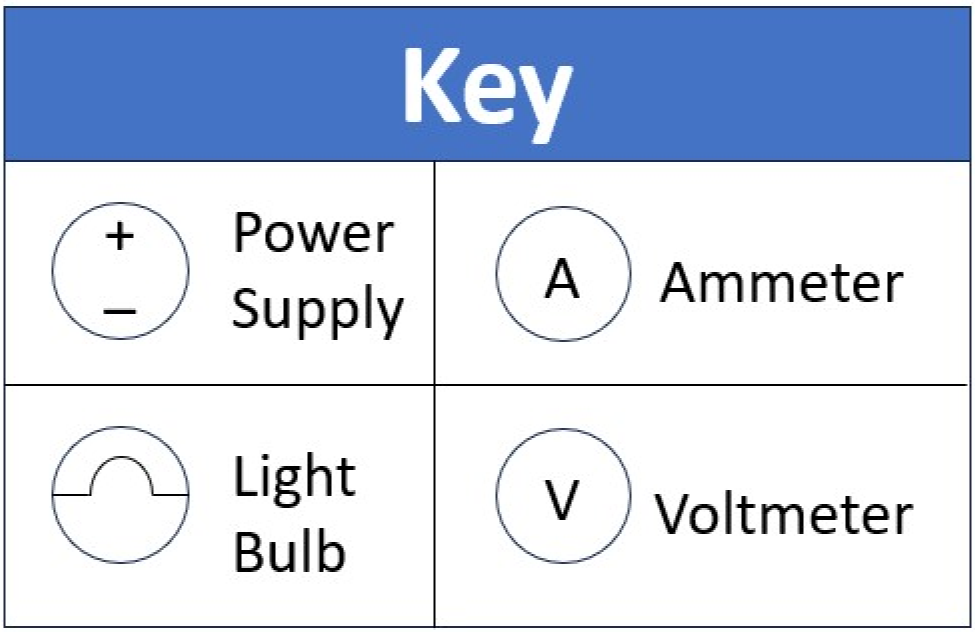
- How do you think adding bulbs to a parallel circuit will affect the brightness of each bulb?
- How do you think it will affect the total current in the circuit?
- How might the voltage measured across each bulb compare to the voltage the power supply provides? Explain your thinking.

11. Select the Parallel Simulation tab on the Series vs. Parallel Circuits spreadsheet.

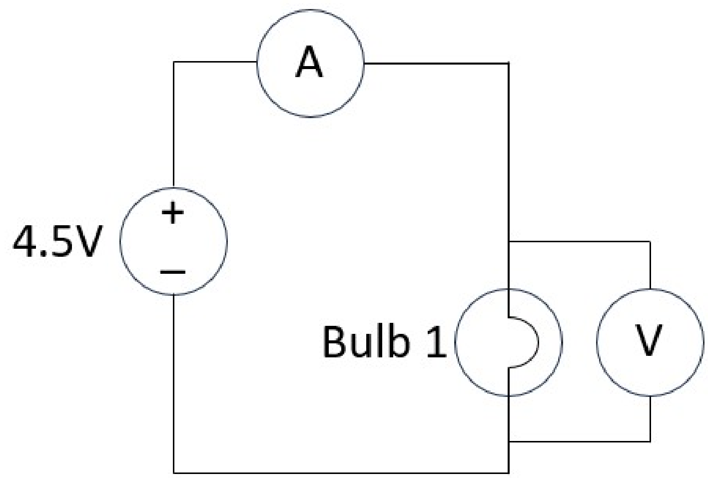
12. Observe the Circuit Schematic Key and Circuit Schematics 1-3. For each one, note the value of the power supply (in volts) and the

locations of multimeters to measure current (A) and voltage (V).

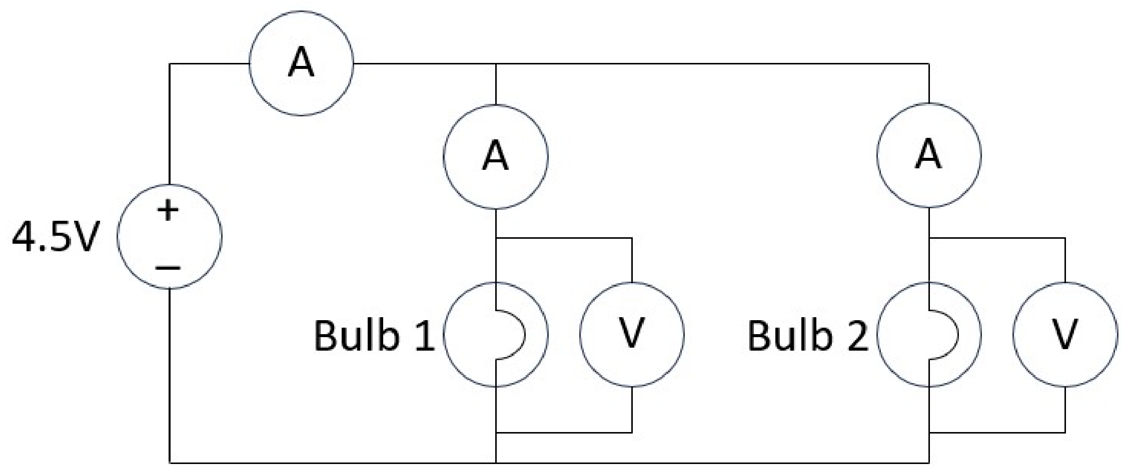
CIRCUIT SCHEMATIC KEY	CIRCUIT SCHEMATIC 1	CIRCUIT SCHEMATIC 2	CIRCUIT SCHEMATIC 3



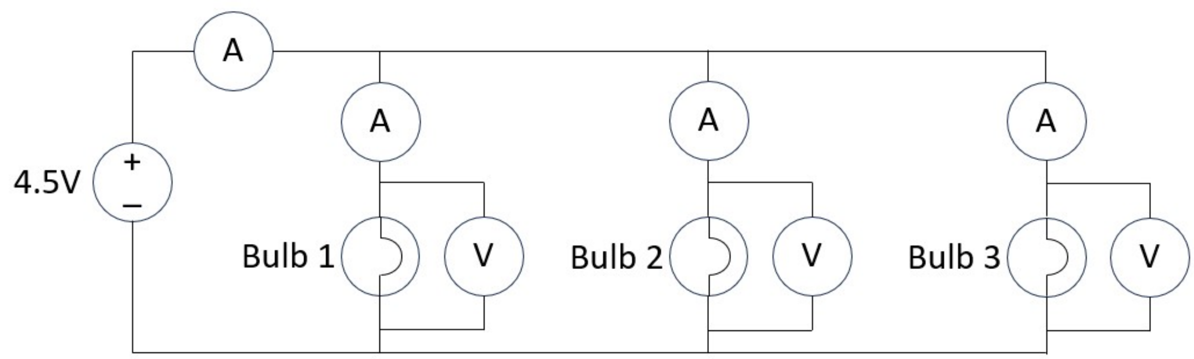
CIRCUIT SCHEMATIC KEY	CIRCUIT SCHEMATIC 1	CIRCUIT SCHEMATIC 2	CIRCUIT SCHEMATIC 3



CIRCUIT SCHEMATIC KEY	CIRCUIT SCHEMATIC 1	CIRCUIT SCHEMATIC 2	CIRCUIT SCHEMATIC 3



CIRCUIT SCHEMATIC KEY	CIRCUIT SCHEMATIC 1	CIRCUIT SCHEMATIC 2	CIRCUIT SCHEMATIC 3



13 Using the Circuit Schematics 1-3 from above, model all three circuits in the same Tinkercad workspace.

14 Start the simulation in Tinkercad. Collect and record the data requested on the spreadsheet.

Reflection

- Referencing your data, how does the number of bulbs in a parallel circuit affect the current in that circuit? How does the number of bulbs affect the brightness of each bulb?
- How does the current you measured before each bulb relate to the total current measured next to the power supply?
- How does the total voltage provided by the power supply relate to the voltage across each bulb? Explain your thinking using data.

Mathematical Relationships for Parallel Circuits

Like you did with series circuits, you'll use the data you collected to develop mathematical relationships for the total current and voltage in a parallel circuit.

15 Describe and record the mathematical relationship between total current (I_T) and current through individual circuit branches (I_1, I_2, I_3, ... I_n) in a parallel circuit.

16 Describe and record the mathematical relationship between total voltage (V_T) and voltage across individual circuit branches (V_1, V_2, V_3, ... V_n) in a parallel circuit.

Time to Practice

Using your mathematical relationships, solve the following practice problems. You may need to algebraically manipulate your mathematical relationship depending on what quantity you are solving for.

17 Determine the voltage across R2 in Figure 4.

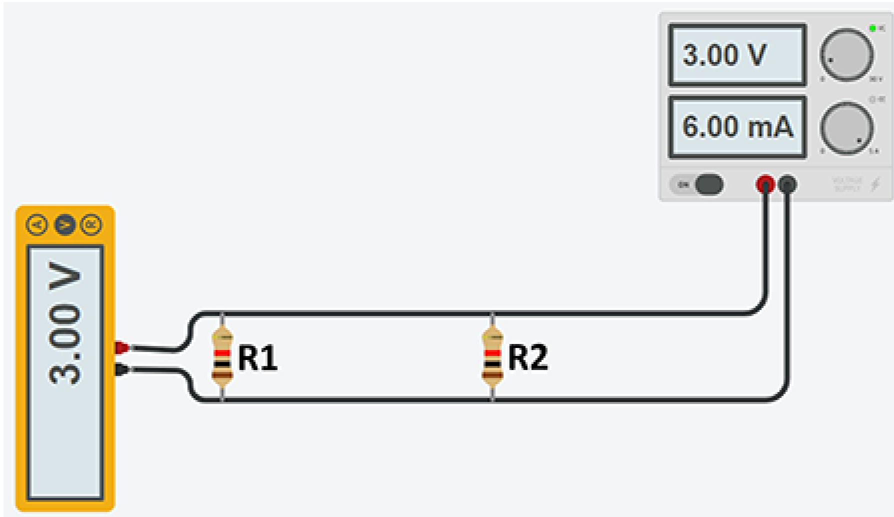


Figure 4. Parallel Circuit with Two Resistors

18 Consider a parallel circuit with two resistors connected to a 6 V power supply with a total current draw of 4 mA. If the current through one resistor is 1.5 mA, determine the current through the second resistor.

19 Determine the current draw of R3 in Figure 5.

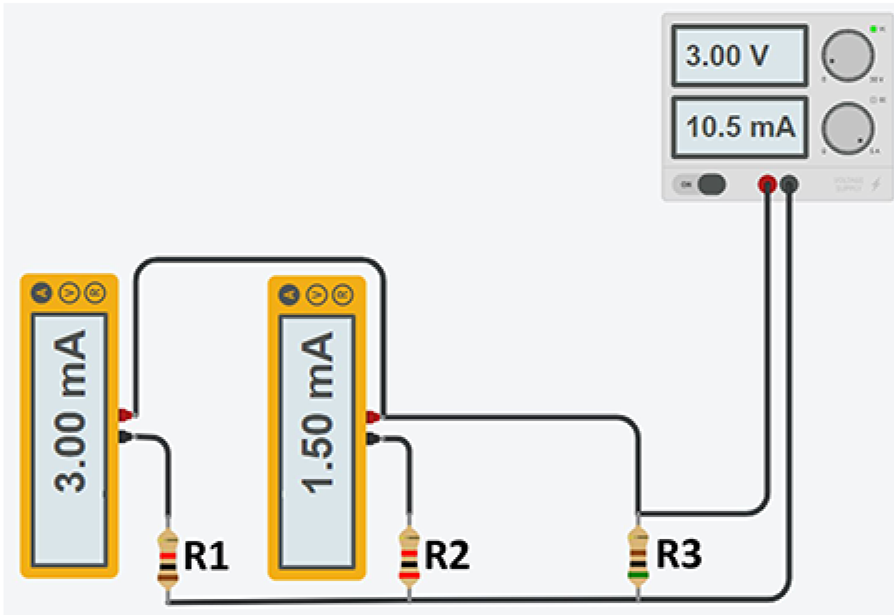


Figure 5. Parallel Circuit with Three Resistors

Wiring and Testing Circuits

Measuring voltage and current in simulated circuits can be an effective way to generate mathematical relationships between circuit values. However, it is important to experimentally verify those rules by creating and testing physical circuits.

Series Circuits

20 Construct a series circuit using a variable power supply set to 3 V, a breadboard, and three resistors (100 Ω, 220 Ω, and 330 Ω).

21 Use the multimeter to measure:

- Total voltage
- Total current
- Voltage across each resistor
- Current before each resistor

Record all data in the Series Circuit tab of the Series vs. Parallel Circuits spreadsheet.

NOTE: In this circuit, for any one resistor, Ohm's Law (or the mathematical relationship between voltage, current, and resistance) applies.

Reflection

Does the data support the mathematical relationship you generated for voltage and current in series circuits? To what can you attribute any variation in your data?

22 Demonstrate that Ohm's Law applies to each resistor in your circuit. Remember to first convert current values from mA to A as needed.

Parallel Circuits

23 Construct a parallel circuit using a variable power supply set to 3 V, a breadboard, and three resistors (100 Ω, 220 Ω, and 330 Ω).

24

Record all data in the Parallel Circuit tab of the Series vs. Parallel spreadsheet. Use the multimeter to measure:

- Total voltage
- Total current
- Voltage across each resistor
- Current before each resistor

NOTE: In this circuit, for any one resistor, Ohm's Law (or the mathematical relationship between voltage, current, and resistance) applies.

Reflection

Does the data support the mathematical relationship you generated for voltage and current in parallel circuits? What can you attribute any variation in your data to?

25 Demonstrate that Ohm's Law applies to each resistor in your circuit. Be sure to first convert current values from mA to A as needed.

Conclusion

Question 1

Determine the current through each resistor of the circuit in Figure 6, and the voltage across each resistor. Explain your reasoning.

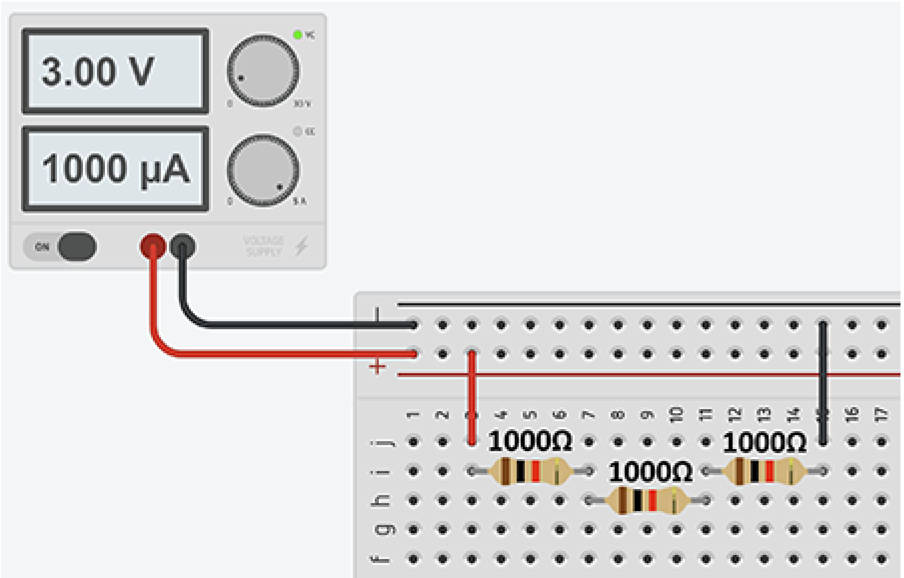


Figure 6. Series Circuit

Question 2

Determine the current through each branch of the circuit in Figure 7, and the voltage across each resistor. Explain your reasoning.

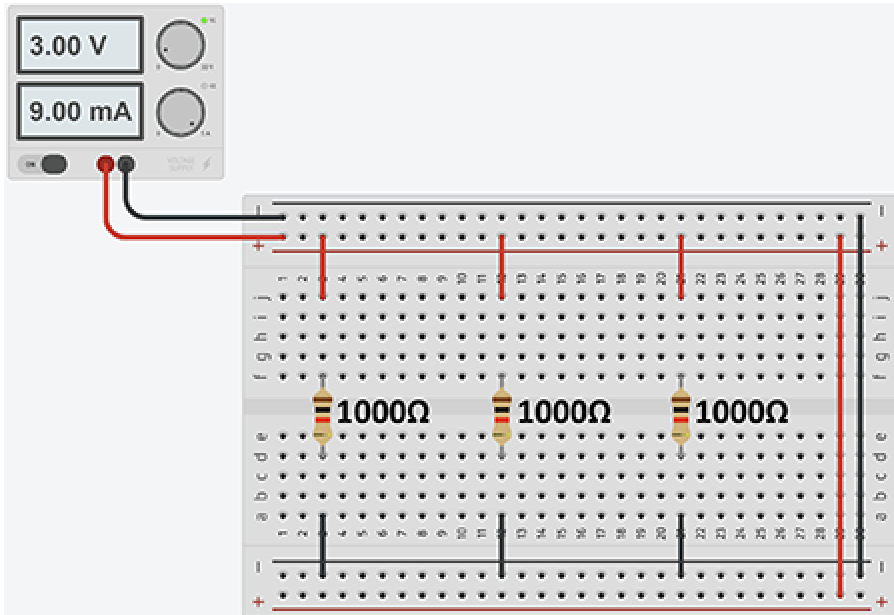


Figure 7. Parallel Circuit